Chapter 14
Digital Holography

Phase measurement is one of the most important problems in Optics. When light waves are scattered by an object, the information about the object is encoded in the scattered waves in the form of amplitude as well as phase variations (apart from possible changes in polarization, spectrum, etc.). The phase information in scattered light waves is often considered to be much more valuable compared to the amplitude information. All detectors operating at visible wavelengths respond to time-averaged energy incident on individual detection element like a pixel. Ordinary photography cameras do manipulate the phase of incoming light waves by means of a lens assembly but only record the intensity (or amplitude) information. Holography is an interferometric technique that records both amplitude as well as phase information. Holographic imaging was first demonstrated by Gabor [1] and a major advance in the form of an off-axis hologram was demonstrated by Leith and Upatnieks [2]. There are several excellent resources that discuss the history and recent developments of holography as provided in references to this chapter [3, 4]. The main idea behind the possibility of holographic imaging was already discussed in the context of Rayleigh-Sommerfeld-Smythe diffraction theory in Chap. 9. In particular we obtained the important result for free-space diffraction that if the tangential E-field is specified in a particular plane (say $z = 0$), then all the components of the E-field are completely determined in the right half space. Consider an object that on illumination by a coherent beam of light produces a certain tangential field $O(x, y, z = 0)$ in the plane $z = 0$ as shown in Fig. 14.1. Now suppose that the same field can be generated in the $z = 0$ plane by some other means so that an observer in the $z > 0$ region receives the same diffracted field as before. In that case even if the original object is not physically present to the left of the screen, the viewer of the scene will perceive its presence. It is clear that the field $O(x, y, z = 0)$ has both amplitude as well as phase in general which needs to be recorded and then re-created. One of the simplest means to generate the same field $O(x, y, z = 0)$ is to record it

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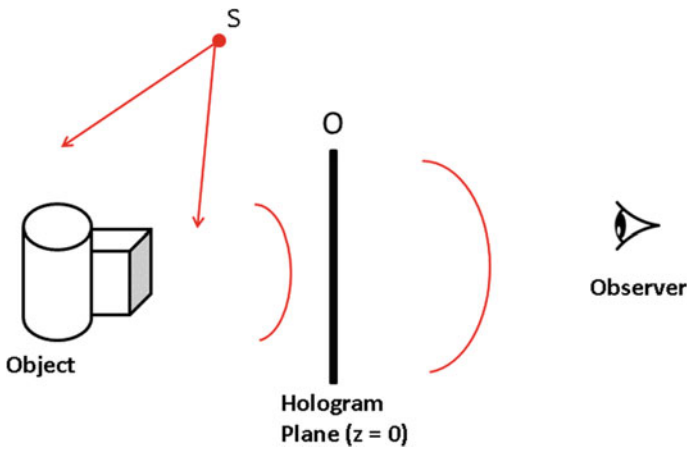


Fig. 14.1 Spatially coherent wavefront from source S illuminates an object to produce scattered field whose tangential scalar component in the hologram plane is given by $O(x, y, 0)$. If this field component can be recorded and generated in the hologram plane again, then the viewer on the right-hand side will get the same perception as if the object was present in its original location

interferometrically. As we have already studied in Chap. 10, an interference pattern with good fringe contrast (or visibility) may be recorded if the two interfering beams are mutually coherent. Typically a reference beam R is derived from the same source by means of a beamsplitter device and is interfered with the object beam O at the $z = 0$ plane which we will now refer to as the hologram plane. If a square-law area detector is placed in plane $z = 0$, the interference pattern H recorded on the detector is described as:

$$H = |R + O|^2 = |R|^2 + |O|^2 + R^* O + R O^*. \tag{14.1}$$

The phase information about the object wave O is recorded in an encoded form in the last two terms of the equation above. The amplitude as well as phase information about the object beam may be recorded this way using multiple configurations. For example, if the field $O(x, y)$ at the hologram plane represents the Fraunhofer or Fresnel diffraction pattern corresponding to the object being imaged, we may term the hologram as a Fourier transform hologram or a Fresnel hologram . If on the other hand the object field $O(x, y)$ corresponds to the image of the object of interest, as may be recorded by a $4F$ system, the hologram is termed as an image plane hologram.

In traditional film-based holography, the hologram pattern H used to be recorded on a high-resolution photographic film. The replay of the hologram is then achieved by physical re-illumination of the hologram with a counter-propagating or conjugate beam R^*. Assuming that the transmission function of the developed film is linearly proportional to the intensity pattern H, the re-illumination process of the hologram with the R^* beam creates a field profile $R^* H$ in the hologram plane:

$$R^{*}H = R^{*}|R|^{2} + R^{*}|O|^{2} + R^{*2}O + |R|^{2}O^{*}. \tag{14.2}$$

The cross terms in the above equation (third and fourth terms on right-hand side) are important as they are proportional to the complex object waves O and O^{*}. These two terms evolve via free-space diffraction and form real and virtual images of the original object. From early days of holography till mid-1990s, holographic imaging continued to be performed preferentially using high- resolution photographic films although hologram recording on an electronic array detector was demonstrated as early as in 1970s [5]. Our goal in this chapter is to concentrate on the topic of digital holography which has become a popular computational phase imaging modality over the last couple of decades. In digital holography, the hologram is recorded using an electronic array detector like CCD or CMOS sensors. Once the hologram frame H is recorded and is available in a numerical form, the hologram replay process cannot be carried out physically but can only be performed using numerical processing methods [6–11]. We will discuss some of the traditional methods for processing digital holograms (holograms recorded on array sensors) that are in-principle similar to the traditional film-based hologram replay. We will also present a brief discussion of some recent developments in using an optimization approach to complex object wave recovery from holograms. The new optimization approach will be seen to overcome the important limitations of the traditional numerical processing methods. Finally, we will point out an important distinction between 3D visual perception in display holography and computational 3D holographic image reconstruction. In particular *it will be shown that the traditional holographic replay does not produce a numerical field distribution identical to the original 3D object*. We will further present a methodology of what we call as the "true 3D" reconstruction. The overall discussion of these advanced topics brings out an important point that digital holography can go much beyond simple mimicking the holographic reconstruction as achieved by traditional film-based methods.

14.1 Some Practical Aspects of Digital Holography Systems

14.1.1 In-line and Off-axis Configurations

Digital holograms can be recorded in two main configurations, viz. in-line and off-axis geometry. The original Gabor hologram was recorded using the in-line configuration shown in Fig. 14.2a. In an in-line holography system, the objects are typically small in size (e.g. particulates, micro-organisms, etc.) that are illuminated by a plane or spherical wavefront which is large in extent compared to the objects. The weak scattering due to the small objects suggests that the resultant field on the detector is a sum of the illuminating beam as well as the scattered field from the objects. The corresponding interference pattern is commonly referred to as in-line hologram and the typical look of the interference pattern for this case is shown in Fig. 14.2b. From the

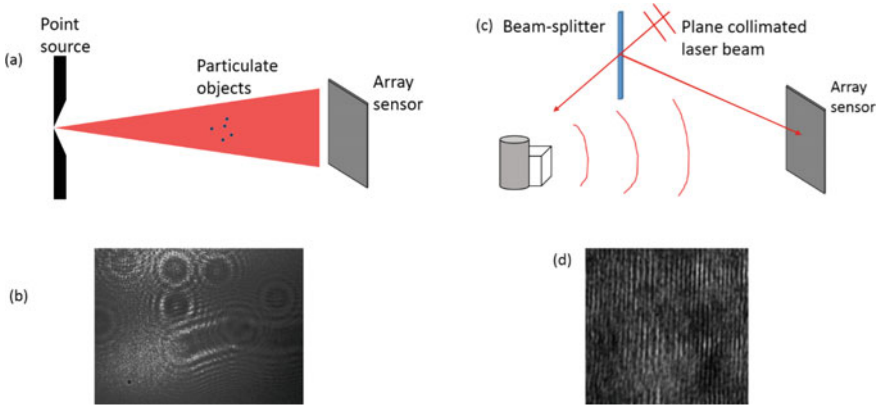


Fig. 14.2 In-line and off-axis holographic recording configurations are shown in **a**, **c**, respectively. The typical holograms recorded with these geometries are shown in **b** and **d**

hologram in Fig. 14.2b, we observe interference fringe patterns due to several point objects and a thin thread-like object (seen as an elongated pattern). For a point object in the path of a spherical wave, the approximate form of the interference pattern may be expressed as

$$H \approx |A \exp[i\frac{\pi}{\lambda z_1}(x^2 + y^2)] + B \exp[i\frac{\pi}{\lambda z_2}((x - x_0)^2 + (y - y_0)^2)]|^2. \quad (14.3)$$

Here z_1, z_2 refers to the distance between the point source and the point object from the array sensor and (x_0, y_0) are the transverse coordinates of the point object. The amplitudes A and B above have a typical relationship $|A| >> |B|$ due to weak scattering amplitude from the point object. Further the amplitudes A, B also fall off inversely with the distance between the point object or point source and the detector coordinate. Referring to Fig. 14.2a, we note that when multiple weak scattering centers are present in the reference beam path, the second term in Eq. (14.3) will be replaced by a summation of individual scattered fields. The resultant interference pattern contains multiple local circular fringe patterns which may or may not overlap. It may be noted that both the reference and object beams travel along common path in in-line configuration. The off-axis hologram configuration on the other hand involves a reference beam which is incident on the detector at an angle as shown in Fig. 14.2c relative to the nominal direction of the object beam. This configuration typically leads to straight line fringes as shown in Fig. 14.2d that are modulated due to the object beam. If the off-axis reference beam is denoted by a tilted plane wave $R = R_0 \exp(i2\pi f_0 x)$ and the object beam at the detector is denoted by $O(x, y)$, the interference pattern recorded on the array sensor may be described as:

$$H = |R|^2 + |O|^2 + 2|R||O| \cos[2\pi f_0 x - \phi_O(x, y)]. \quad (14.4)$$

Here $\phi_O(x, y)$ represents the phase map associated with the complex-valued object wave at the detector plane. The fringes described as per the above equation are shown in Fig. 14.2d.

14.1.2 Sampling of Digital Holograms

When recording an interference pattern on an array sensor, it is important to remember that the oscillations in the interference fringe pattern need to be sampled appropriately at least as much as governed by the Nyquist sampling rate. If the angle between the k-vectors of two plane waves (derived from the same source) is given by θ as shown in Fig. 14.3, then the straight line interference fringes associated with this configuration have a period Λ given by

$$\Lambda = \frac{\lambda}{2 \sin \theta}. \tag{14.5}$$

Here λ is the illuminating wavelength. Clearly *as per the Nyquist criterion, we must have at least two samples of the interference pattern per fringe period along every pixel row which is normal to the fringe direction*. In practice it is suitable to have 4−5 pixels span one fringe period since any array sensor does not really measure point samples but rather integrates light irradiance over the pixel size. For sensor pixel size of 5 μm and $\lambda = 0.5\,\mu$m, the nominal angle between two beams cannot be very large (at most 1−2 degrees) when recording a digital hologram. In this respect, digital holography differs from film-based holography as high-resolution films readily allow much larger angle (30−40 degrees) between two interfering beams. For circular fringes as in in-line holography similar considerations apply so that the largest visible circular fringe is sufficiently sampled. In the discussion hereafter we will assume that holograms have been recorded with the appropriate sampling considerations.

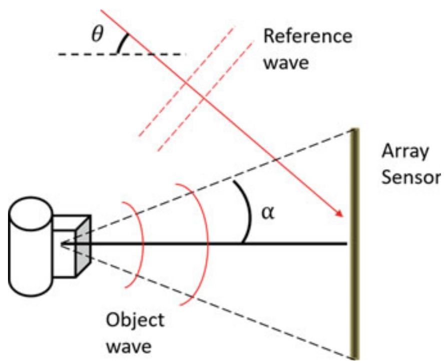


Fig. 14.3 Numerical aperture for the hologram recording geometry. Numerical aperture limited by detector size is given by $NA = \sin \alpha$

14.1.3 Numerical Aperture of the Hologram Recording System

In both in-line and off-axis configurations shown in Fig. 14.2, it is important to understand that the numerical aperture ($NA = \sin\alpha$) subtended by the detector with respect to the nominal object position decides the extent of angular spectrum from the scattered object field received at the sensor (see Fig. 14.3). When we discuss the recovery of complex-valued object field in the hologram plane or consider the holographic reconstruction or replay process the effective NA plays an important role in deciding the recoverable information (in terms of lateral and longitudinal resolution) about the test object to be imaged. In particular, one may expect a lateral resolution of $\sim \lambda/(NA)$ and a longitudinal resolution of $\sim \lambda/(NA)^2$ from a configuration as shown in Fig. 14.3.

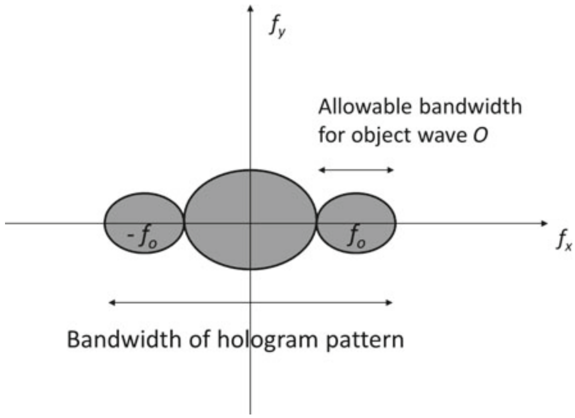
14.2 Complex Object Wave Recovery in the Hologram Plane

In this section, we will discuss three methods for recovery of complex object wave $O(x, y)$ from the recorded hologram(s) $H(x, y)$ as in Eq. (14.1). Following this, in the next section, we will examine the question of image reconstruction using the recovered object field $O(x, y)$.

14.2.1 Off-axis Digital Holography

Off-axis digital holography that was first proposed in the work of Leith and Upatnieks [2] uses a plane wave reference beam at a nominal angle θ relative to the object wave $O(x, y)$ (see Fig. 14.3). The reference beam may be described as $R = R_0 \exp(i2\pi f_0 x)$ where $f_0 = \sin\theta/\lambda$ and the resulting interference pattern can be described as in Eq. (14.4). The amplitude $|R|$ of the plane reference beam is essentially constant (or have very low frequency features in an experiment) across the hologram. The Fourier transform structure of the off-axis hologram is as shown in Fig. 14.4 and consists of three main regions where the energy in the hologram signal is concentrated. The first two term $|R|^2$ and $|O(x, y)|^2$ are located near the center of the (f_x, f_y) plane and are known as the dc terms. The cosine term in Eq. 14.4 can be split into two terms that are located at spatial frequency locations $(f_0, 0)$ and $(-f_0, 0)$. The information of interest about the complex object wave $O(x, y)$ is contained in one of the side lobes in the Fourier plane. The simplest approach to recover this information is to filter out the side-lobe by numerical processing. For this method to work, one must make sure that the three different lobes as in Fig. 14.4 do not overlap with each other substantially. Suppose the expected bandwidth of the

Fig. 14.4 Fourier transform representation of the off-axis hologram showing dc and cross terms



object wave is $2B$ along the x-direction, the width of the central lobe in the Fourier transform plane corresponding to the term $|O(x, y)|^2$ is $4B$ and the transform of $|R|^2$ is almost a delta function located at the zero frequency. In order to make sure that the dc and the cross terms remain separated, we must therefore have:

$$f_0 = \frac{\sin\theta}{\lambda} \geq 3B. \tag{14.6}$$

The reference beam angle is thus selected carefully to avoid overlap of terms in Fourier space and subsequent loss of information about the object wave. We observe that the allowable object wave bandwidth is much smaller than the total hologram bandwidth along the x-direction due to the non-overlap condition. In single-shot operation, one may process the digitally recorded hologram signal in following procedure which is commonly referred to as the Fourier transform method (FTM).

1. Compute the 2D Fourier transform of the off-axis hologram.
2. Locate the carrier-frequency peak (or centroid) and filter out appropriate region of the Fourier transform plane centered on it.
3. Shift the filtered region to zero frequency (or dc) in Fourier transform plane.
4. Compute inverse Fourier transform of the dc shifted field to obtain the complex object field $O(x, y)$ in the hologram plane.

This procedure was first described by Wolf [12] and later by Takeda [13]. In the step 2 above, locating the carrier-frequency peak in the spatial frequency domain is not always a straightforward matter. The finite resolution of the FFT function used to compute the Fourier transform of the off-axis hologram may make it difficult to locate the carrier-frequency peak accurately. If the hologram is for example represented on 2D array of size $N \times N$, the spatial frequency resolution on using the FFT function is $\Delta f_x = \Delta f_y = 1/N$. There is no guarantee that the peak due to the carrier frequency will fall at the center of a pixel in the Fourier transform domain. If the carrier-fringe

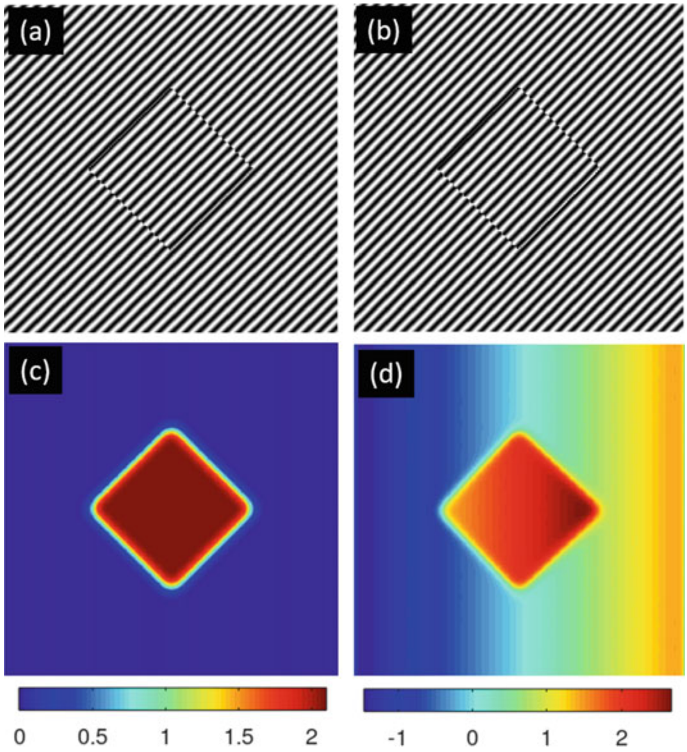


Fig. 14.5 Phase map obtained by following the nominal four step procedure for the Fourier transform method. **a**, **b**: Two holograms of the same step phase object where the fringe directions are slightly rotated with respect to each other, **c**, **d**: Phase maps show relative tilt phase due to sub-pixel shift in the carrier-frequency peak

frequency shifts this peak location by Δ_x, Δ_y, respectively ($|\Delta_x| < 0.5$ pixel and $|\Delta_y| < 0.5$ pixel), the resultant solution may show an overall tilt phase given by

$$\Delta\phi(x, y) = \frac{2\pi}{N}(\Delta_x x + \Delta_y y). \tag{14.7}$$

As an illustration in Fig. 14.5, we show two holograms for a square-shaped binary step phase object. The phase step in this illustration is equal to $2\pi/3$. The fringes in the two holograms look nearly same to the eye but have a slight relative rotation between them. If the four step procedure outline above is followed, we observe that the resultant phase map has a tilt phase background in one of the phase recoveries as shown in Fig. 14.5c, d. This tilt has appeared despite the fact that the object wave in both the holograms is identical. In order to address this fractional fringe detection issue, the pixels near the carrier-frequency peak need to be sampled at higher sampling density to estimate any sub-pixel peak shift. This is best achieved by evaluating the local discrete Fourier transform (DFT) of the hologram near the carrier-frequency peak [14]. After evaluating the carrier-frequency peak (u_0, v_0) from the FFT of the hologram, we may thus proceed as follows: A local DFT centered on (u_0, v_0) may be computed as a product of three matrices:

$$F(\mathbf{U}, \mathbf{V}) = \exp(-i\frac{2\pi}{N}\mathbf{U}\mathbf{X}^T)\; H(\mathbf{X}, \mathbf{Y})\; \exp(-i\frac{2\pi}{N}\mathbf{Y}\mathbf{V}^T). \quad (14.8)$$

Here $\mathbf{X}, \mathbf{Y}$ are spatial coordinate vectors given by $\mathbf{X} = \mathbf{Y} = [-N/2, -N/2+1, ..., N/2-1]^T$ (assuming even N) in the hologram plane. The spatial frequency coordinates $(\mathbf{U}, \mathbf{V})$ are centered on (u_0, v_0). If we want to sample a neighborhood of say 1.5 pixels surrounding (u_0, v_0), this may be achieved by defining:

$$\begin{aligned} U &= u_0 + [-\frac{1.5}{2}, -\frac{1.5}{2} + \frac{1}{\alpha}, -\frac{1.5}{2} + \frac{2}{\alpha}, ..., \frac{1.5}{2} - \frac{1}{\alpha}]^T \\ V &= v_0 + [-\frac{1.5}{2}, -\frac{1.5}{2} + \frac{1}{\alpha}, -\frac{1.5}{2} + \frac{2}{\alpha}, ..., \frac{1.5}{2} - \frac{1}{\alpha}]^T. \end{aligned} \quad (14.9)$$

In the above relations, α is the up-sampling factor with which we wish to sub-sample the pixels near the carrier-frequency peak. For a hologram of size $N \times N$, this process matrix product has three matrices of size $(1.5\alpha \times N)$, $(N \times N)$ and $(N \times 1.5\alpha)$ which is marginal additional computation when $\alpha << N$. Typically size of the holograms is $N \sim 10^3$, whereas an up-sampling factor $\alpha \sim 10^1$ is sufficient. For the case illustrated in Fig. 14.5, the magnitude of the three matrix product in two cases for $\alpha = 20$ is shown in Fig. 14.6. The two matrices are of size 30×30 and represent the sub-sampled representation of 1.5 pixel near the pixel (u_0, v_0) in Fourier transform of the hologram. The sub-pixel shift corresponding to the tilt phase in Fig. 14.5d is clearly visible in Fig. 14.6b. This tilt can be corrected by finding the sub-pixel peak shifts Δ_x, Δ_y as in Eq. (14.7) from the matrix in Fig. 14.6b and subtracting the resultant $\Delta\phi(x, y)$ from the computed phase map in Fig. 14.5b. This sub-pixel peak shift correction is important for standardizing the usage of off-axis interferometry systems. The phase recovery result using the Fourier transform methodology as illustrated in Fig. 14.5 suggests that even after the sub-pixel shift correction, the methodology still has one drawback that the spatial resolution of the recovered phase

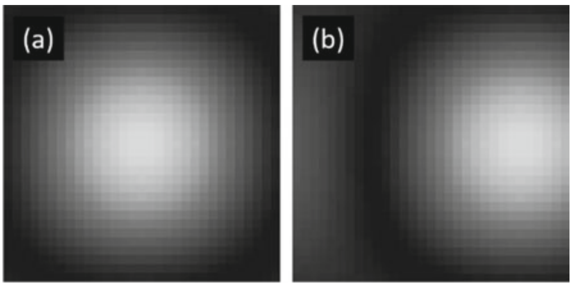


Fig. 14.6 **a**, **b** Show the magnitude of the three matrix product in Eq. (14.8) for the interference patterns in Fig. 14.5a, b, respectively

map is poor due to the low-pass filtering nature of this methodology which does not allow one to reconstruct the sharp features in the phase map. Note that this resolution limitation is purely due to the Fourier filtering methodology. It may be observed from Fig. 14.5 that the interference data itself contains the information about sharp edges in the form of fringe discontinuities. As we will discuss in the next sections, the lost resolution can be regained by a multi-step phase shifting methodology or by employing a more advanced optimization methodology for complex object wave reconstruction from a single interference data frame.

14.2.2 Phase Shifting Digital Holography

Phase shifting is a technique [15] may be used to regain the lost resolution in Fourier transform method. Multiple digital holograms are recorded on a digital array sensor with known phase shifts in the reference beam. For example if the R beam gets four pre-determined phase shifts $0, \pi/2, \pi$, and $3\pi/2$, the corresponding digital holograms as per Eq. (14.1) are given by

$$\begin{aligned}
H_1 &= |R|^2 + |O|^2 + 2|R||O|\cos(\phi_O - \phi_R), \\
H_2 &= |R|^2 + |O|^2 - 2|R||O|\sin(\phi_O - \phi_R), \\
H_3 &= |R|^2 + |O|^2 - 2|R||O|\cos(\phi_O - \phi_R), \\
H_4 &= |R|^2 + |O|^2 + 2|R||O|\sin(\phi_O - \phi_R).
\end{aligned} \tag{14.10}$$

In the above equation, the terms ϕ_O and ϕ_R denote the 2D phase functions corresponding to the object and reference beams, respectively, at the hologram plane. Known phase shifts can be added to the reference beam in several ways, e.g. introducing appropriate combination of wave retarder plates in the reference beam, introducing a QHQ (Q=quarter wave plate, H= half wave plate) geometric phase shifter in the reference beam, using a piezo-electric transducer to shift the location of an optical component like a mirror in the reference beam path, etc. By adding the known reference phase shifts to the reference beam, it is now possible to obtain both the sine and cosine quadratures of the unknown phase. If the reference phase ϕ_R is known, we can obtain the object beam phase as

$$\phi_O(x, y) = \phi_R(x, y) + \arctan(\frac{H_4 - H_2}{H_1 - H_3}). \tag{14.11}$$

Observe that unlike off-axis holography, the reference beam does not have to be an off-axis plane wave but can in general have any known phase profile. In Fig. 14.7 we diagrammatically show the complex object recovery for a step phase object (step size equal to $\pi/2$) using the phase shifting method. For illustration a spherical reference wave is used. The steps in the recovery of the square shaped step phase object are illustrated in Fig. 14.7 and we clearly see the phase object recovery with full

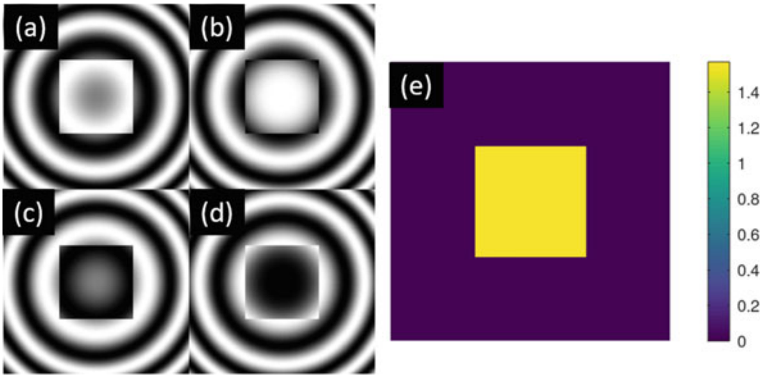


Fig. 14.7 Phase shifting digital holography, **a**–**d**: Four holograms for the square step phase object with reference beam phase shifts given by 0, $\pi/2$, π, $3\pi/2$, respectively. **e** Recovery of object phase using Eq. (14.11)

pixel resolution. At the cost of recording multiple frames the phase shifting digital holography regains the full detector resolution. It is required that during the recording of the multiple hologram frames, the object remains stationary. Also it is important to remember that while recording the four data frames as in Eq. (14.10) in the laboratory, each data frame contains noise (e.g. detector readout noise and photon noise), and as a result by processing the four frames as above there is a noise penalty to pay in terms of accuracy of the measured phase.

14.2.2.1 Accuracy Analysis of Classical Interferometers

The analysis for phase measurement accuracy is typically performed by analyzing the measurements at the two output ports of an interferometer as shown in Fig. 14.8. The unknown phase is assumed to be $(\pi/2 + \Delta\phi)$ for convenience. If N photons are incident on the interferometer as shown in Fig. 14.8, the detectors D_1 and D_2 (assumed to be ideal) at the two output ports of the interferometer will detect average number of photons $N_{1,2}$ given by

$$N_{1,2} = \frac{N}{2}[1 \pm \sin(\Delta\phi)]. \tag{14.12}$$

From the phase measurement perspective the signal of interest is

$$N_s = N_1 - N_2 = N \sin \Delta\phi \approx N\Delta\phi, \tag{14.13}$$

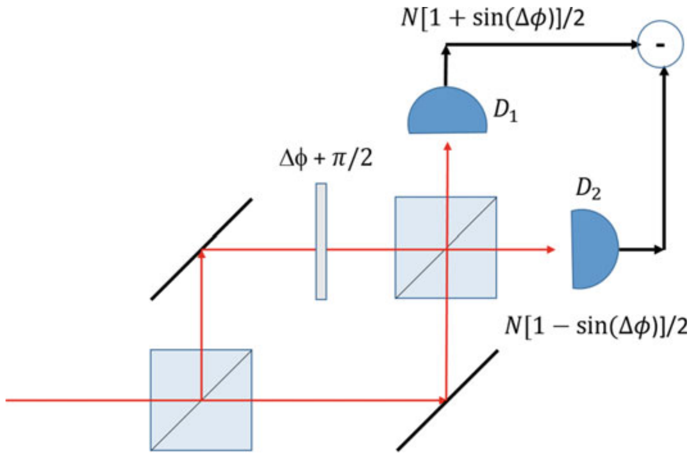


Fig. 14.8 Accuracy of phase measurement in an interferometer

for small angle $\Delta\phi$. For classical light sources and typical detection integration times, the photon counts at the two detectors $N_{1,2}$ follow Posisson statistics. The standard deviation in the signal N_s can then be determined as

$$\sigma_{N_s} = \sqrt{\sigma_{N_1}^2 + \sigma_{N_2}^2} = \sqrt{N}. \tag{14.14}$$

The signal-to-noise ratio for phase detection is therefore equal to $N_s/\sigma_{N_s} = \sqrt{N}\Delta\phi$. Assuming that the lowest detectability limit occurs when the SNR is equal to 1, we observe that the phase measurement accuracy is given by $1/\sqrt{N}$. The accuracy limit is commonly referred to as the shot-noise-limit (SNL) or the standard quantum limit. A number of efforts are underway to improve this phase measurement accuracy for highly sensitive interferometer devices like the advanced gravitational wave interferometer. The typically proposed approaches for this purpose involve adding squeezed vacuum state through the second input port of the interferometer. We however note that the whole accuracy analysis above is focused on single detected pixel of the interference signal at the two symmetrically placed point detectors. Measured phase signals are seldom single pixel in nature and as soon as the phase signal has a multi-pixel nature (in space or time), ideas such as image sparsity that we discussed in the first part of the book immediately become applicable. Accuracy improvement beyond SNL is then possible even with classical light interferometers. It is just that now the phase recovery must happen via an optimization algorithm that uses appropriate sparsity constraints. We will discuss this optimization approach to phase recovery from an interferogram in detail next.

14.2.3 Optimization Method for Complex Object Wave Recovery from Digital Holography

The Fourier transform method and the phase shifting method for complex object wave reconstruction are well-established, however, they have certain limitations due to the nature in which these methods approach the problem of information recovery from a digital holograms. For example, while Fourier transform method can perform single-shot holographic imaging, it by default achieves lower resolution than what the digital detector array used for recording the hologram is capable of. Due to sampling considerations, the allowable reference beam angle for off-axis hologram is already quite small and the non-overlap requirement for the dc and cross-term lobes in Fourier domain further reduces the attainable object resolution in this configuration. The phase shifting method regains the detector resolution, however, multiple hologram frames have to be recorded. Further, in terms of photon resources, the phase shifting method has to divide the total number detected photons into four camera records (for arbitrary phase values in $[0, 2\pi]$) and the SNR for each interference record is therefore poor. The optical system has to be stable during the multiple recordings and further the object needs to be stationary during the recording process. High-quality vibration isolation is not always possible in field environment where digital holographic systems may have to be deployed. Single-shot holographic imaging systems have a distinct practical advantage in this regard. The optimization approach discussed in the following is promising in that it can recover phase images with full pixel resolution even with a single-shot operation.

We start by writing the hologram equation Eq. (14.1) in a slightly modified form.

$$|O|^2 + 2|O||R|\cos(\phi_O - \phi_R) + (|R|^2 - H) = 0, \qquad (14.15)$$

where $\phi_O(x, y)$ and $\phi_R(x, y)$ are the phase maps associated with the object and reference beams, respectively. For every pixel (x, y), the above equation is a quadratic in $|O|$ and therefore we can write a solution:

$$|O| = -|R|\cos(\phi_O - \phi_R) \pm \sqrt{H - |R|^2 \sin^2(\phi_O - \phi_R)}. \qquad (14.16)$$

At each pixel location (x, y), one may choose the positive valued solution as we are solving for $|O|$. From the above solution, we note that $|O|$ is dependent on the phase ϕ_O even when the complex-valued reference beam is known via a calibration procedure. Any arbitrary unknown phase function ϕ_O along with the solution $|O|$ can therefore be used to construct a complex object wave which satisfies Eq. (14.1). The performance of any practical device which uses interference phenomenon depends on how well one can recover the complex object wave information. The ambiguity in the solution as discussed above is not desirable and a procedure for arriving at a meaningful solution must be discussed carefully as we will do next.

We have already discussed in Chapter 6 that an appropriate method for image reconstruction in presence of such ambiguities is to formulate the problem as a constrained optimization problem [16–18]. In particular we consider the problem of minimization of a cost function of the form:

$$C(O, O^*) = ||H - (|R|^2 + |O|^2 + R^*O + RO^*)||^2 + \alpha\,\psi(O, O^*). \quad (14.17)$$

The first term above is the least square or L2-norm squared error that represents consistency of the solution O with the hologram data H. The reference beam is assumed to be known as is the case with the Fourier transform method or the phase shifting method. The second term $\psi(O, O^*)$ is the constraint term that models some desired property of the solution O depending on the Physics of the problem. For example, if we are recording a Fresnel hologram, we expect the solution for object wave $O(x, y)$ to have a smooth profile whereas in case of an image plane hologram, the solution $O(x, y)$ is expected to retain information about sharp edges in the object to be imaged. The optimization problem above is peculiar in that the cost function to be minimized is real and positive valued but the solution of interest $O(x, y)$ is complex-valued. For optimization purpose, it is necessary to evaluate the functional gradient of the cost function to find the descent direction in which the solution is to be progressed starting with some initial guess. We have already studied one such problem in Chap. 6 and recall that direction of steepest descent for δC is obtained when the variation δO is in the direction of $\nabla_{O^*}C$. The simplest gradient descent iteration for obtaining the solution $O(x, y)$ may now be written as

$$O^{(n+1)} = O^{(n)} - t[\nabla_{O^*}C]_{O=O^{(n)}}, \quad (14.18)$$

where $O^{(n)}$ denotes the guess solution for O at the nth iteration. The Wirtinger derivative $[\nabla_{O^*}C]$ may be evaluated as

$$\nabla_{O^*}C = -2[H - |R + O|^2] \cdot (R + O) - \alpha\nabla_{O^*}\,\psi. \quad (14.19)$$

The step size t may be selected using standard backtracking line search. Further the image recovery may also be performed by methodologies discussed in Section 6.5, so that, no free parameter α in required for object wave recovery. In the illustration in Fig. 14.9, we first show the effect of α on the phase solution $\arg(O)$. In this numerical example, both $|R|$ and $|O|$ are constants and equal to 1. The same off-axis hologram as in Fig. 14.5a has been used for the iterative reconstruction. Phase map reconstruction for three values of $\alpha = 0.1, 1, 10$ is shown. For penalty function we use the total variation (TV) penalty. From the reconstructions, the solution for regularization parameter $\alpha = 1$ appears to be the most appropriate for a step phase object. For $\alpha = 0.1$, the regularization is not enough and the fringe-like artifacts are present in the phase map. For $\alpha = 10$ there is over-regularization and the solution appears to loose the sharp edge feature of the step phase object. The illustration for $\alpha = 1$ however shows that sharp edge recovery with much higher resolution compared to the Fourier filtering method (see, e.g. Figure 14.5b) is possible via optimization approach. *Full pixel resolution in single-shot should thus be possible to achieve if the complex object wave recovery is formulated as an optimization*

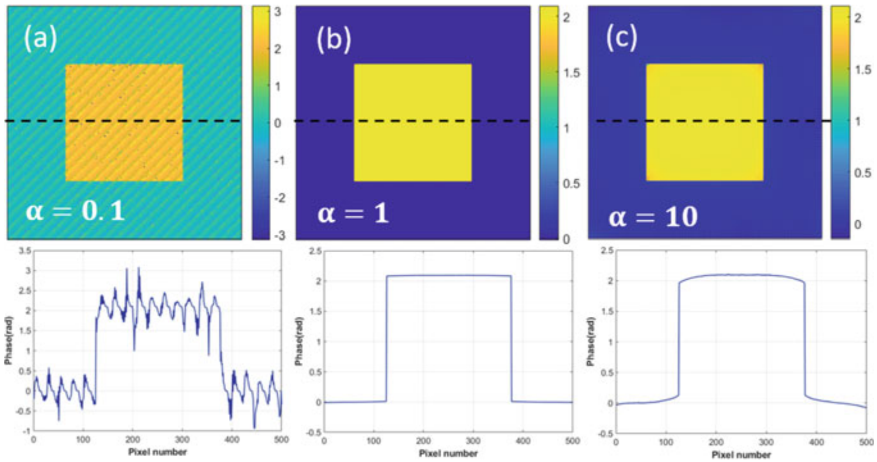


Fig. 14.9 Iterative reconstruction of phase map from single-shot off-axis interferogram. Reconstructions with three different values of regularization parameter $\alpha = 0.1, 1, 10$ are shown in **a**, **b**, **c**, respectively. The profiles of the phase recoveries along the dotted line are also plotted

problem. Similar to the discussion in Section 6.5, the selection of value of α is a tedious process in this example as well. The mean gradient descent (MGD) approach as discussed in the context of the de-convolution can also be applied in the present problem. In Fig. 14.10, we show phase reconstruction for an off-axis as well as on-axis interferogram using the MGD approach [19]. The sharp phase object feature can be reconstructed as seen in Fig. 14.10c, d from single-shot interferogram at full pixel resolution. Interestingly, the optimization procedure can recover the object wave for on-axis case as well, for which there is no solution otherwise except for using a multi-shot phase shifting method. We would however caution the readers that if the fringe density is too low as in Fig. 14.7, the penalties like TV which only consider the next nearest neighbor in their definition may not be the most appropriate for suppressing fringe artifacts and overall the algorithm may take longer to converge to appropriate solution.

14.2.4 Noise Advantage Offered by the Optimization Method

The accuracy of measured phase is very important for sensitive interferometric systems. While we have discussed this topic in Sect. 14.2.2.1, the traditional treatment presented there uses a single-pixel-based methodology for studying accuracy of classical interferometers. One of the ideas that has been explored in detail in the literature is to use the light states that offer sub-Poisson statistics of photon stream for improving phase accuracy beyond the shot noise limit. While single-pixel-based analysis of noise is followed in most texts, we wish to point out that phase signals of inter-

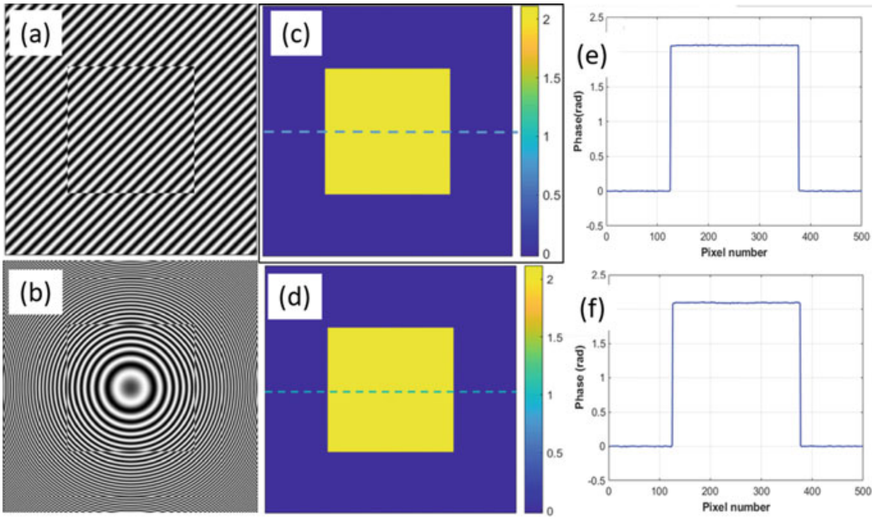


Fig. 14.10 Iterative reconstruction of phase map from single-shot off-axis as well as on-axis interferograms using MGD approach which does not require any regularization parameter α. Note that optimization method can in this case handle on-axis case as well. The profiles of the phase recoveries along the dotted line are plotted. Adapted with permission from [19] ©Optica

est are almost never single-pixel in practice. One may be interested in measuring a time-domain waveform at a single-pixel or in measuring a 2D phase front as is often the case in digital holography experiments. In such cases, the signal of interest will almost always have some structure. The in-built redundancy or sparsity of the phase signal to be reconstructed can then be utilized via an optimization algorithm to obtain RMS phase error that is better than the single-pixel-based shot noise limit. In the numerical illustrations shown in Fig. 14.10, the interferograms have been simulated with Poisson noise and the recovered RMS phase errors are observed to have error performance better than SNL. The noise advantage of utilizing the sparsity of phase map in the reconstruction algorithm can best be appreciated at low light levels where shot noise becomes prominent. In Fig. 14.11b we illustrate an experimental measurement of phase map due to a convex lens object [20]. The lens object is placed in one arm of a Mach-Zehnder interferometer arrangement. The QHQ phase shifter (see Chap. 11) is introduced in reference arm of the interferometer in order to introduce four phase shifts in the reference beam. The interference pattern is recorded on an EMCCD (Electron Multiplying CCD) array sensor which can operate in a photon counting mode. When four interferograms are recorded with high light level ($>$ 5000 photons/pixel), the four interferograms processed using Eq. (14.11) provide a smooth ground truth phase map as shown in Fig. 14.11a. In Fig. 14.11c–e we show three interferograms in the same arrangement recorded by reducing the detected light level to 200, 50, 10 photons/pixel, respectively. In the second and third columns in Fig. 14.11c–e, we show phase reconstruction results obtained using four-step phase

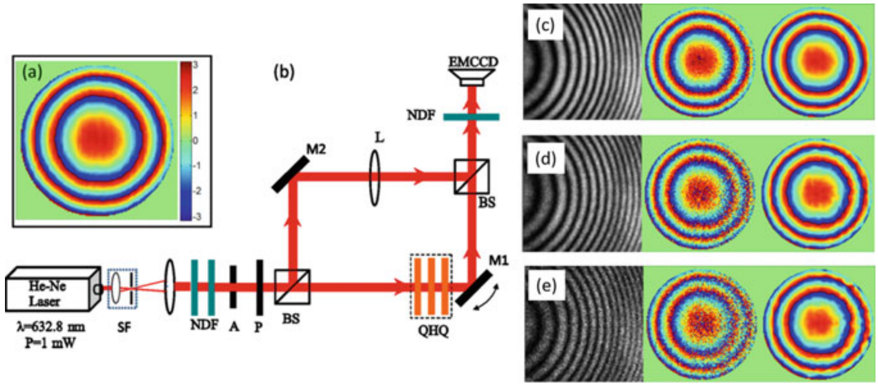


Fig. 14.11 Sub-shot-noise phase measurement using sparsity-based phase retrieval. **a**, **b** Show ground truth phase map for a lens object when measured using a Mach-Zehnder interferometer. **c**–**e** Typical interferograms at low light levels with 200, 50 and 10 photons/pixel. Phase reconstructions using four-step phase shifting method and single-shot optimization method are shown in two columns adjoining the interferograms. Photon budget for the two methods is kept equal. The pixel-to-pixel phase variation is not present in the optimization solution. Adapted with permission from [20] ©American Physical Society

shifting method and the optimization method, respectively. The total photon budget in the two methodologies is same, meaning that same photon flux is divided into four frames for the phase shifting method. Here the penalty function used was a sum of the (weighted) squared difference function of the form:

$$\psi(O, O^*) = \sum_{i=all\ pixels} \sum_{p,q \in window} w_{pq} \, |O_{i,p} - O_{i,q}|^2, \tag{14.20}$$

as this penalty is appropriate for a smooth object like a lens. In the above equation $O_{i,p}$ denotes pth pixel in a window centered on i-the pixel in the intereference record. The weight function $w_{p,q}$ is inversely proportional to the distance between pixels p, q in a window of 5×5 pixels. We observe that the phase shifting method operates individually on a pixel to pixel basis (see Eq. (14.11) whereas in the optimization method the neighboring pixels interact with each other through the penalty function. As a result, when the light level is reduced, the phase shifting solution gets noisy whereas the optimization solution still remains smooth and similar to the ground truth solution. In this experiment [20], it was observed that at low light level of 10 photons/pixel in the interferogram data, the RMS phase accuracy of the recovered phase map for the optimization solution with respect to the ground truth phase map was more than 5-fold better than the SNL-based phase accuracy which is observed to hold for the phase shifting method. This result indicates the importance of signal sparsity in essentially beating the traditional limits like SNL for realistic applications where the phase map to be measured is often expected to have sparsity. Finally, we note that the sparsity ideas are independent of the nature of light source used and therefore can further enhance performance of an interferometer that uses a non-classical light state with sub-Poisson photon statistics.

14.3 Digital Holographic Microscopy

The ability of measurement of quantitative phase information has found important applications in microscopic imaging of label-free or unstained cells [21]. Most light microscopy in bio-science laboratories or pathology clinics is currently performed using intensity-based imaging modalities such as bright-field or fluorescence. The contrast mechanism in these methods are dyes used for staining of cells or fluorescence agents that are used to tag particular parts of a cell. While these methodologies have been popular, they involve wet lab processing of cells that affects their natural state. Quantitative phase imaging offers an attractive alternative that uses the natural refractive index of cells as a contrast mechanism. When a plane laser beam is transmitted through a transparent cell, under weak scattering assumption, the phase of the transmitted beam is given by

$$\phi(x, y) = \frac{2\pi}{\lambda} \int dz\, n(x, y, z). \tag{14.21}$$

As depicted in Fig. 14.12, a plane wave passing through a cell sample undergoes a spatially dependent phase change as in this equation. Here $n(x, y, z)$ stands for the relative refractive index of the cell with respect to the surrounding medium, which is in general a function of coordinates (x, y, z). The microscope objectives and the tube lens combine as depicted in Fig. 14.12 make a 4F-like (infinity corrected) imaging system in both the arms of the interferometer which is a transmission version of the Linnik type interferometer. A nominal image plane hologram for blood cell sample is shown for illustration. The off-axis hologram recorded in this balanced Mach-Zehnder interferometer configuration consists of straight line fringes which show bending at the location of cells indicating a phase modulation. With the prior discussion in this chapter, it is clear that the single-shot optimization method offers important advantage in terms of resolution as well as phase accuracy. We note that the optimization method fully operates in the image domain and as a result can also be used for region-of-interest (ROI) reconstruction from an image plane hologram. Phase reconstruction of an individual red blood cell is also shown in Fig. 14.12 as a surface plot where the color coding refers to the optical path difference experienced by the transmitted laser beam. The optimization method used for this illustration has used Huber penalty (see Problem 6.2). More details on the topic of ROI phase reconstruction from an image plane hologram can be found in [22]. This depth information is not available with commonly used bright-field microscopes. We note that the contrast provided by bright-field microscopes is due to absorption of light in stained cells. On the other hand, the quantitative phase information $\phi(x, y)$ as in Eq.

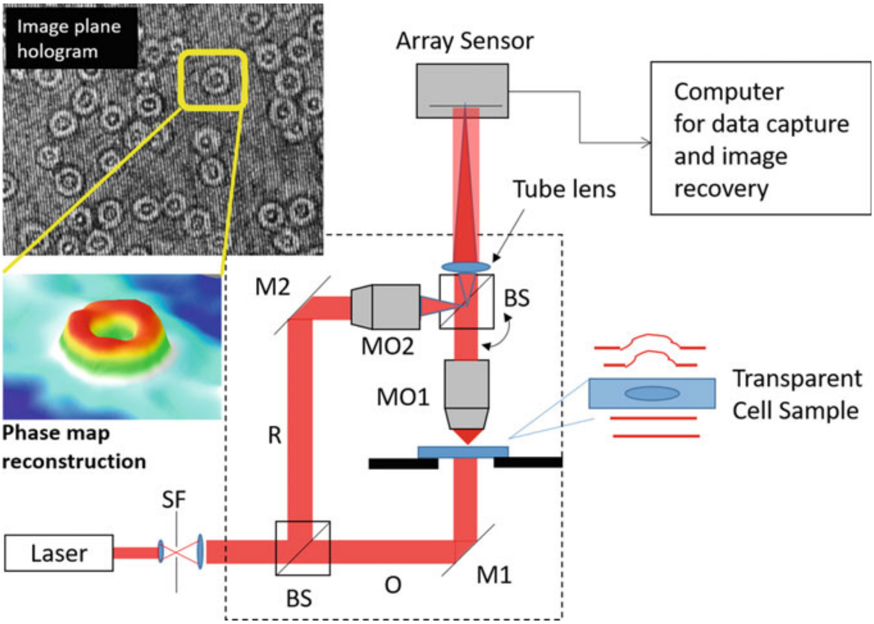


Fig. 14.12 A typical balanced digital holographic microscope (DHM) arrangement employing a Mach-Zehnder arrangement is shown, SF: spatial filter, M1, M2:mirrors, MO1, MO2: Microscope objectives, BS: Beamsplitter, R: Reference beam, O: Object beam. The curved arrow near the second beamsplitter shows how R beam may be tilted to introduce off-axis beam angle. Adapted with permission from [22] ©Taylor & Francis Group

(14.21) as provided by a DHM instrument depends on the locally varying refractive index of cells which depends on their chemical properties. DHM methodology is therefore useful in studying small refractive index changes that may happen in cells leading to phase changes. The accurately measured phase change can in fact serve as an important parameter for cell image-based pathological diagnosis [23]. While this methodology is not yet commonly used in clinics, the sensitive phase imaging methodology has great potential for future application development. From basic biosciences perspective, DHM offers an important modality for studying live cells in their natural state as no staining or fluorescent labeling of cells is required in this methodology.

Several variants of DHM system depicted in Fig. 14.12 have been explored in the literature. It is possible to use unbalanced interferometer systems with a single microscope objective in the object beam arm. Such a system design however requires compensation of aberrations due to the microscope objective from the recovered phase map [27, 28]. Compact 3D printed or mobile phone camera-based DHM system designs have also been studied by now and offer attractive possibility of low cost point-of-care diagnostic devices. It is important to remember that apart from phase reconstruction, a practical DHM system also requires additional algorithms for auto-focusing, phase unwrapping, etc. to make a complete system. A useful phase unwrapping algorithm based on the transport of intensity equation has already been discussed in Sect. 9.5.1 which can be readily used with a DHM system. The auto-focusing in digital holography involves propagating the complex object field $O(x, y)$ to nearby planes where the definition of object features may be sharper thus correcting for any de-focus error in recording of the hologram. One of the powerful criteria for this purpose is to use the sparsity-of-gradient criterion. In particular, the recovered object field $O(x, y)$ may be propagated to various z distances as follows:

$$O_z(x, y) = O(x, y) * h_{AS}(x, y, z), \tag{14.22}$$

where $h_{AS}(x, y, z)$ is the angular spectrum impulse response (computation above is preferably performed using transfer function and FFTs). A gradient magnitude image $G = |\nabla_\perp O_z(x, y)|$ is computed and a sparsity measure like the Tamura coefficient (TC) defined as

$$TC = \sqrt{\frac{\sigma_G}{\langle G \rangle}}, \tag{14.23}$$

is then used to determine the best focus plane [26]. In the above equation, σ_G and $\langle G \rangle$ stand for the standard deviation and the mean of the gradient magnitude image G, respectively. In the best focus plane, the TC measure is usually observed to show a distinct peak. For unstained phase objects, best focus plane can also be located simply by observing the image plane hologram [29] as the sample is de-focused sequentially through focus. At the best focus plane the fringes in the hologram predominantly show phase modulation and minimal amplitude modulation. Such a technique can be used for deciding the focus plane while recording the hologram and provides a similar user experience like that with a bright-field microscope, particularly for bio-science researchers. We will discuss the topic of auto-focusing further in the context of in-line holography of particulates in the next section.

14.4 In-line Digital Holography of Particulates or Weak Scattering Objects

In-line digital holography can be a very useful tool when localization and tracking of particulates or other small objects like micro-organisms in a liquid are of interest. The interesting feature of an in-line holography system is its compact nature. Further, since the illumination and the scattered wave essentially follow the same path, holography can be performed with a source of somewhat lower temporal coherence. Given a recorded in-line hologram as shown in Fig. 14.2b, the main task for reconstruction problem is to propagate the hologram back to the original object volume. This may for example be performed using the angular spectrum propagation approach explained in Chap. 9. A back-propagation of the recorded hologram to two different distances

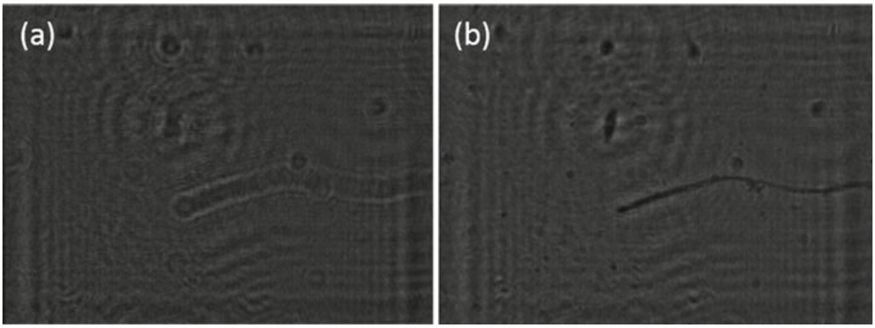


Fig. 14.13 **a**, **b** Computed intensity pattern obtained by numerical back-propagation of the in-line hologram shown in Fig. 14.2b to two different distances

is shown in Fig. 14.13. We clearly observe that in Fig. 14.13b, a number of features (a thread-like elongated objects and point-like objects) are sharp and focused. While propagating the inline hologram back, one needs to take into account the system magnification given by $M = (z_1 + z_2)/z_1$, where z_1 is the source to object distance and z_2 is the object to sensor distance. If an object is originally located at a nominal distance of z_2 from the sensor, then on back-propagation, the best focus distance for that object will be Mz_2. Depending on the known bounding box where the sample is located (e.g. size of a cuvette), the back-propagation process can be limited to a range of distances. For opaque objects, sometimes a minimum intensity criterion is found to be useful for determining the best focus plane. The distance z_2 can be determined by applying focusing criterion like the Tamura coefficient discussed in previous section (see Eq. (14.23)) for automating the focus detection process. A number of auto-focusing criteria in in-line digital holography have been studied comparatively in the literature [30]. We wish to mention here that while the small weak scattering objects do appear to focus in the simple back-propagation-based reconstruction as shown in Fig. 14.13b, this reconstruction is actually corrupted by the twin image which is inevitable in the back-propagation process. This twin image can be eliminated if the reconstruction problem is formulated as an optimization problem employing object sparsity [31, 32]. We will discuss this topic more in the next Section where we discuss the possibility of a tomographic reconstruction in digital holography for sparse objects.

14.5 True 3D Image Reconstruction in Digital Holography

The early work in holography was mostly conducted using high-resolution photographic films for recording of holograms. Holographic image reconstruction was also mainly performed for the purpose of display holography or visual inspection by illuminating the recorded holographic film with the conjugate reference beam as described in Eq. (14.2). In the modern era of digital holography the hologram is recorded on a digital array sensor and therefore the image reconstruction has to be performed numerically. The typical procedure of 3D image reconstruction in this case is to follow a two step procedure. The first step involves recovering the complex object wave $O(x, y)$ in the hologram recording plane. This field may then be back-propagated to original object volume as done in the previous section on in-line holography. It may be noted that unlike the back-propagation of in-line hologram, the back-propagation of the object wave $O(x, y)$ does not give rise to any twin image. The nature of the 3D reconstruction however needs more discussion.

We may examine the 3D reconstruction problem from the point of view of degrees of freedom in the 3D volume and the recovered object field $O(x, y)$ in the hologram plane. It is clear that recording of a hologram is a process of transferring information from a 3D volume onto a 2D detector. In that process we now have lesser number of pixel values in the detector plane compared to the number of 3D voxels in the object volume/ bounding box. As discussed in Sect. 14.2, the 2D complex-valued object field $O(x, y)$ is recovered from the recorded hologram(s) in the detector plane with full pixel resolution. Now the process of forming 3D image as in the traditional holographic replay is to simply back-propagate this field to the original object volume which is a linear operation. It is therefore surprising as to how one is able to generate more information in a 3D volume in the holographic replay process. *An important question to ask is whether this holographic replay (or back-propagation) of the numerically recovered object field $O(x, y)$ truly constitutes an inverse operation leading to a 3D reconstruction.* It may be emphasized here that the perception of focused object in physical hologram replay process is aided by focusing capability of observer's eyes that are absent when we are simply reconstructing the 3D object by numerical back-propagation of the object field from hologram plane to the original object volume.

We will consider a Fresnel zone hologram for discussion and introduce a simple model for total object field $O(x, y)$ that may result from a 3D object as shown in Fig. 14.14. Assuming a reflective object, we may divide the object function $U(x', y', z')$ into multiple slices along z-axis. The total object field may then be described by propagating field in the individual slice up to the hologram plane and then integrating all the propagated fields to get the object field $O(x, y)$. Here we have neglected secondary scattering from the object which is a reasonable approximation for the present purpose and has been used in holography traditionally [33]. The total object field may be expressed by an operator $\hat{A}$ acting on the 3D object function $U(x', y', z')$:

$$
\begin{aligned}
O(x, y) &= \hat{A}U \\
&= \int dz' e^{ik(z'-z_d)} \iint dx'dy'\, U(x', y', z')\, h_F(x - x', y - y', z_d - z').
\end{aligned}
\tag{14.24}
$$

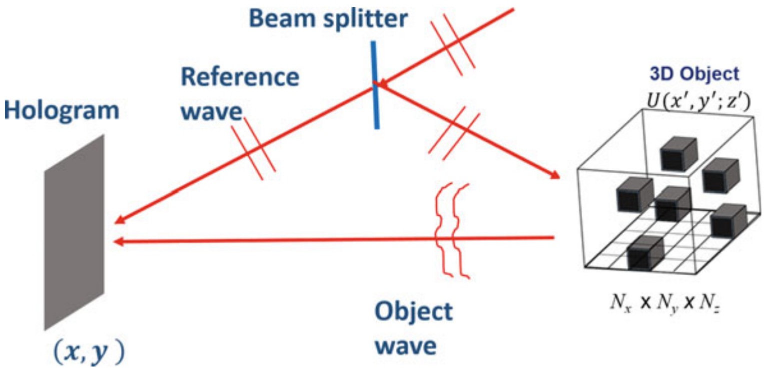


Fig. 14.14 Schematic explaining a simple model for hologram formation due to a 3D object described by $U(x', y', z')$. Adapted from [25] ©Institute of Physics, UK

Here, $h_F(x, y, z)$ stands for Fresnel propagator, z_d refers to the z-coordinate of the detector or the hologram recording plane. The additional phase factor $\exp[ik(z' - z_d)]$ arises since the illuminating wave has different phases for individual object planes. Further we would like to point out that the integral over z' is to be evaluated over the extent of the bounding box containing the 3D object. The interference of this total object field $O(x, y)$ and a reference field $R(x, y)$ is recorded. However the aim of the numerical processing methods in digital holography is to recover the object field $O(x, y)$ from the recorded hologram pattern. The accurately determined object field may therefore be treated as a starting point for our further analysis. Note that physical hologram replay involves using the recorded hologram pattern and therefore the replay field has dc and twin image terms. If the starting point is the numerically recovered object field $O(x, y)$, then the dc or twin terms do not arise in the computed back-propagated field. Next we may determine the adjoint operator $\hat{A}^\dagger$ corresponding to the forward model in Eq. (14.24) by utilizing the scalar product identity:

$$\langle\, O_2,\ \hat{A}U_1 \,\rangle_{2D} = \langle\, \hat{A}^\dagger O_2,\ U_1 \,\rangle_{3D}. \tag{14.25}$$

In the above equation, $O_2(x, y)$ and $U_1(x, y, z)$ refer to an arbitrary 2D object field (at the detector) and a 3D object function, respectively. The suffixes $2D$ and $3D$ on the scalar products $\langle .., .. \rangle$ refer to the fact that the scalar products have been evaluated on the 2D detector coordinates or the 3D volume coordinates. Writing out the above scalar product relation explicitly and rearranging the order of integration leads to the following interesting result:

$$[\hat{A}^\dagger O](x', y', z') = e^{-ik(z'-z_2)} \iint dxdy\, O(x, y)\, h_F^*(x - x', y - y', z_d - z'). \tag{14.26}$$

Apart from the phase factor in front, the above expression for the adjoint operator is precisely the back-propagation of the object field $O(x, y)$ to the original object volume as achieved by traditional holographic replay. We therefore conclude that *the traditional holographic replay corresponds to Hermitian transpose corresponding to the forward hologram formation model* in Eq. (14.24). The numerical back-propagation of $O(x, y)$ therefore does not truly reconstruct the original 3D object but provides an approximation to it. When physically replaying a hologram, the generation of $[\hat{A}^{\dagger} O]$ as an approximation to $U(x', y', z')$ is inevitable, however, in numerical 3D reconstruction, more choices may be available. The 3D reconstruction may for example be formulated as a problem of minimization of the cost function:

$$C(U, U^{*}) = ||O - \hat{A}U||_2^2 + \alpha\, \psi(U, U^{*}). \tag{14.27}$$

As in Chapter 6, the constraint function could for example represent sparsity of the 3D object function $U(x', y', z')$. In the illustration next, we illustrate the 3D reconstruction via optimization as above for an object consisting of four letters A, B, C, D (see Fig. 14.15a) which is a sparse object in the 3D volume of interest. A hologram of this object is recorded and the corresponding object field magnitude on back-propagation of the object wave to the object volume is shown in Fig. 14.15b. As expected, as the object wave propagates from hologram plane back to the object volume, focusing effect is observed at the original location of the letters A, B, C, D. But since the four letters were in different planes initially, back-propagated field in any given plane now consists of a focused field (if an object was present in that plane) along with the de-focused background due to objects in the other planes. Human observers naturally pay more attention to the focused fields and ignore the de-focused background. But from a numerical standpoint we see that the reconstructed field is quite different from the original object field as in Fig. 14.15a. In Fig. 14.15c, we show the reconstruction of the 3D object field that results from solution of the optimization problem

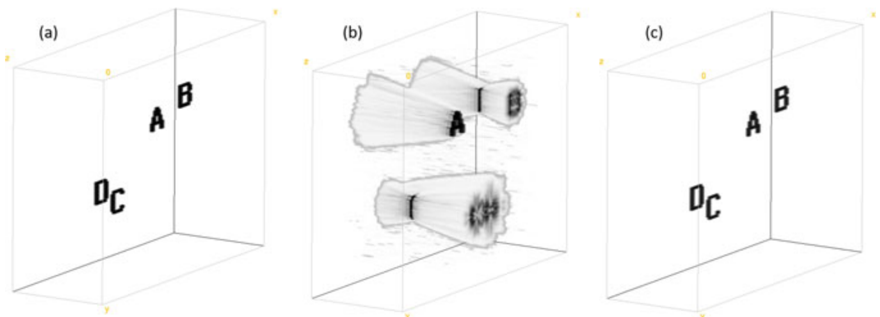


Fig. 14.15 **a** Ground truth 3D object, **b** Reconstructed field magnitude by simple back-propagation of the object field $O(x, y)$ from hologram plane back to the original object volume, **c** Reconstruction of the 3D image via solving a minimization problem for cost function in Eq. (14.27). Adapted from [25] ©Institute of Physics, UK

in Eq. (14.27). This “true 3D” reconstruction is much superior as the object field now predominantly exists only where the letters A, B, C, D were originally located. In this illustration a total variation penalty has been used for encouraging a sparse solution. More details on this illustration may be found in [25]. We emphasize that a CAD-model like 3D reconstruction as observed in Fig. 14.15c is not possible with any physical hologram replay process in which the reconstruction field cannot possibly vanish at slices where no object was originally present. Such reconstruction is thus possible only with digital holography. Digital holographic 3D image reconstruction does not necessarily have to mimic the physical hologram replay process as is a common practice. Instead optimization approach can inherently provide a superior 3D solution. Such true 3D reconstruction may not however be possible for non-sparse objects from degrees-of-freedom considerations. In such cases, additional diversity mechanisms may be required for measuring non-redundant data. The question of whether true 3D reconstruction of a general object is possible is however not just dependent on degrees-of-freedom considerations. The assumption about small secondary scattering that has been used here may not be valid as the 3D object gets more and more complex in its structure. Nevertheless the iterative approach presented here may be valid for 3D reconstruction of low complexity objects such as transparent cells or optical elements like lenses, etc. from their 2D holograms/interferograms. Finally while it is somewhat out of scope in the present discussion, the question of how many holographic measurements are required for an object of given sparsity is not answered fully in our opinion, although partial answers are available based on the literature on diffraction tomography or what is also sometimes referred to as holo-tomography [34].

In summary, this chapter considered the problem of holographic image recording by means of interference phenomena. A number of methods for recovering complex object field from single (or multiple) hologram(s) were discussed with attention to spatial resolution and phase measurement accuracy offered by them. Applications of digital holography to microscopy were considered. Finally some interesting questions regarding possibility of a true 3D/tomographic image reconstruction from hologram data was also considered briefly.

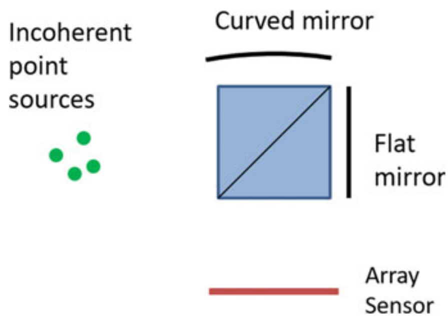


Fig. 14.16 System for holography with incoherent light

Problems

14.1 Holography with spatially incoherent illumination Consider multiple incoherent point sources as in a fluorescence microscopy sample (Fig. 14.16). The radiation from these sources is passed through a Michelson interferometer system which has a flat mirror in one arm and a curved mirror in another arm such that the path lengths along the optic axis are matched. Obtain an expression for the pattern formed on the array sensor at the output port of the interferometer. Simulate the pattern by assuming reasonable system geometry. Explain how this hologram pattern can be utilized to form a tomographic image of the original object. For an interesting twist on how multiple measurements of this nature can be treated as a single hologram with spatially coherent illumination, refer to [24].

14.2 Fringe projection profilometry: Tactile diagrams are commonly used for teaching visually impaired children. These diagrams are printed on a specialized polymer sheet such that the features protrude out of the sheet and can be understood by touch. In order to measure height profile of the tactile diagram, a sinusoidal fringe pattern is projected onto the tactile map surface and the surface is then imaged (see Fig. 14.17). Explain what computational and calibration steps will be required to recover the feature height information.

14.3 Spherical waves from an on-axis point source located at $z = 0$ and an in-line plane wave traveling in the $+z$-direction (assume that the two sources are derived from the same laser beam and thus have required coherence properties) are used to form a hologram in plane $z = z_0$. The hologram was recorded using illumination wavelength λ_1 nm but for reconstruction purpose, a plane wave traveling in negative z direction and having a wavelength λ_2 nm is used. Obtain an expression for the hologram intensity pattern and simulate the pattern in computer. Find the location of the reconstructed real image.

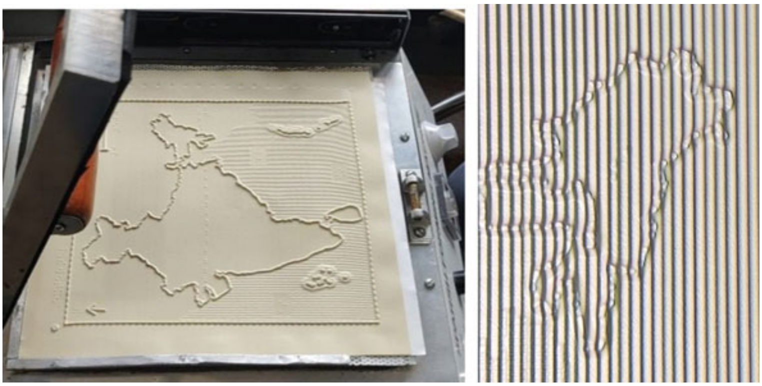


Fig. 14.17 Tactile diagram of a map and part of map with fringe projection

14.4 Fringe demodulation using Hilbert transform: Consider an off-axis hologram with straight line fringes oriented at an arbitrary angle in the camera record. The low frequency terms in the hologram may be estimated by low-pass filtering of the recorded fringe pattern and removed. Describe how the spiral phase transform (using the $\exp(i\phi)$ or spiral phase filter in the Fourier domain) is helpful for fringe demodulation process.

14.5 Traditionally an in-line hologram is processed by simple back-propagation as illustrated in Fig. 14.13. This method however has a problem that the reconstructed object consists of a background field associated with the intensity and twin terms in the hologram. Formulate an iterative optimization procedure in order to get rid of these extra terms so that the reconstruction of the object in the appropriate focus plane is quantitatively meaningful.

References

1. D. Gabor, A new microscopic principle. Nature **161**, 177–178 (1948)
2. E.N. Leith, J. Upatnieks, Reconstructed wavefronts and communication theory. J. Opt. Soc. Am. **52**, 1123–1128 (1962)
3. M.K. Kim, Principles and techniques of digital holographic microcopy. SPIE Rev. **1**, 018005-1–018005-50 (2010)
4. B. Javidi, A. Carnicer, A. Anand, G. Barbastathis, W. Chen, P. Ferraro, J.W. Goodman, R. Horisaki, K. Khare, M. Kujawinska, R.A. Leitgeb, P. Marquet, T. Nomura, A. Ozcan, Y. Park, G. Pedrini, P. Picart, J. Rosen, G. Saavedra, N.T. Shaked, A. Stern, E. Tajahuerce, L. Tian, G. Wetzstein, M. Yamaguchi, Roadmap on digital holography [invited]. Opt. Express **29**, 35078–35118 (2021)
5. J.W. Goodman, R.W. Lawrence, Digital image formation from electronically detected holograms. Appl. Phys. Lett. **11**, 77–79 (1967)
6. H. Chen, M. Shih, E. Arons, E.N. Leith, J. Lopez, D. Dilworth, P.C. Sun, Electronic holographic imaging through living human tissue. Appl. Opt. **33**, 3630–3632 (1994)
7. U. Schnars, W.P. Juptner, Direct reconstruction of holograms by a CCD target and numerical reconstruction. Appl. Opt. **33**, 179–181 (1994)
8. E. Cuche, F. Bevilacqua, C. Depeursinge, Digital holography for quantitative phase contrast imaging. Opt. Lett. **24**, 291–293 (1999)
9. E. Cuche, P. Marquet, C. Depeursinge, Simultaneous amplitude-contrast and quantitative phase-contrast microscopy by numerical reconstruction of Fresnel off-axis holograms. Appl. Opt. **38**, 6994–7001 (1999)
10. M. Liebling, T. Blu, M. Unser, Complex-wave retrieval from a single off-axis hologram. J. Opt. Soc. Am. A **21**, 367–377 (2004)
11. L. Yaroslavsky, *Digital Holography and Digital Image Processing* (Kluwer Academic, 2010)
12. E. Wolf, Determination of the amplitude and the phase of scattered fields by holography. J. Opt. Soc. Am. A **60**, 18–20 (1970)
13. M. Takeda, H. Ina, S. Kobayashi, Fourier transform method of fringe pattern analysis for computer based topography and interferometry. J. Opt. Soc. Am. **72**, 156–160 (1982)
14. M. Singh, K. Khare, Accurate efficient carrier estimation for single shot digital holographic imaging. Opt. Lett. **41**, 4871–4874 (2016)
15. I. Yamaguchi, T. Zhang, Phase shifting digital holography. Opt. Lett. **22**, 1268–1270 (1997)

16. K. Khare, P.T.S. Ali, J. Joseph, Single shot high resolution digital holography. Opt. Express **21**, 2581–2591 (2013)
17. P.T.S. Ali, K. Khare, J. Joesph, Quantitative phase imaging with single shot digital holography. Opt. Commun. **319**, 85–89 (2014)
18. M. Singh, K. Khare, Single-shot interferogram analysis for accurate reconstruction of step phase objects. J. Opt. Soc. Am. A **34**, 349–355 (2017)
19. S, Rajora, M. Butola, K. Khare, Mean gradient descent: an optimization approach for single-shot interferogram analysis. J. Opt. Soc. Am. A **36**, D7–D13 (2019)
20. M. Singh, K. Khare, A.K. Jha, S. Prabhakar, R.P. Singh, Accurate multipixel phase measurement with classical light interferometry. Phys. Rev. A **91**, 021802(R) (2015)
21. Y.K. Park, C. Depeursing, G. Propescu, Quantitative phase imaging in biomedicine. Nat. Photon. **12**, 578 (2018)
22. M. Singh, K. Khare, Single-shot full resolution region-of-interest (ROI) reconstruction in image plane digital holographic microscopy. J. Mod. Opt. **65**, 1127–1134 (2018)
23. J. Mangal, R. Monga, R. Mathur, A.K. Dinda, J. Joseph, S. Ahlawat, K. Khare, Unsupervised organization of cervical cells using bright-field and single-shot digital holographic microscopy. J. Biophotonics **12**, e201800409 (2019)
24. J. Rosen, G. Brooker, Digital spatially incoherent Fresnel holography. Opt. Lett. **32**, 912–914 (2007)
25. J. Birdi, S. Rajora, M. Butola, K. Khare, True 3D reconstruction in digital holography. J. Phys. Photonics **2**, 044004 (2020)
26. Y. Zhang, H. Wang, Y. Wu, M. Tamamitsu, A. Ozcan, Edge sparsity criterion for robust holographic autofocusing. Opt. Lett. **42**, 3824–3827 (2017)
27. E. Sánchez-Ortiga, A.I. Doblas, G. Saavedra, M. Martínez-Corral, J. Garcia-Sucerquia, Off-axis digital holographic microscopy: practical design parameters for operating at diffraction limit. Appl. Opt. **53**(10), 2058–2066 (2014)
28. A.I. Doblas, E. Sánchez-Ortiga, M. Martínez-Corral, G. Saavedra, J. Garcia-Sucerquia, Accurate single-shot quantitative phase imaging of biological specimens with tele-centric digital holographic microscopy. J. Biomed. Opt. **19**, 046022 (2014)
29. R. Malik, P. Sharma, S. Poulose, S. Ahlawat, K. Khare, A practical criterion for focusing of unstained cell samples using a digital holographic microscope. J. Microsc. **279**, 114–122 (2020)
30. T. Zimmerman, A. Park, S. Bianco, Evaluating autofocusing metrics for plankton holographic microscope image reconstruction. Proc. SPIE **11710**, 117100A (2021)
31. F. Soulez, L. Denis, C. Fournier, É. Thiébaut, C. Goepfert, Inverse-problem approach for particle digital holography: accurate location based on local optimization. J. Opt. Soc. Am. A **24**, 1164–1171 (2007)
32. L. Denis, D. Lorenz, E. Thiébaut, C. Fournier, D. Trede, Inline hologram reconstruction with sparsity constraints. Opt. Lett. **34**, 3475–3477 (2009)
33. D. Gabor, Theory of electron interference experiments. Rev. Mod. Phys. **28**, 260 (1956)
34. V. Balasubramani, A. Kuś, H.Y. Tu, C.J. Cheng, M. Baczewska, W. Krauze, M. Kujawińska, Holographic tomography: techniques and biomedical applications. Appl. Opt. **60**, B65–B80 (2021)